Generative Image Modeling on BloodMNIST Dataset Using VAE, DCGAN, and Diffusion Models

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Abstract. Generative modeling plays a vital role in machine learning, enabling systems to learn complex data distributions and synthesize novel samples indistinguishable from real data. In this study, we investigate three prominent generative architectures — Variational Autoencoders (VAE), Deep Convolutional Generative Adversarial Networks (DCGAN), and Conditional Diffusion Models — applied to the BloodMNIST dataset, which consists of microscopic images of peripheral blood cells. We implement, train, and evaluate these models on the dataset, using the Fréchet Inception Distance (FID) as the primary quantitative metric to assess image quality and diversity. Our findings demonstrate that DCGAN achieves the best generative performance with the lowest average FID score of 96.88, surpassing the diffusion model's 108.46 and the VAE's 172.70. These results underscore the trade-offs between sample fidelity, training complexity, and latent space representation inherent in different generative approaches, highlighting the efficacy of adversarial training for realistic biomedical image synthesis.

Keywords: VAE, DC-GAN, Diffusion Models, FID

Introduction

Recent advances in generative modeling have significantly impacted various fields by facilitating the synthesis of novel data samples that closely mimic the statistical properties of original datasets. This capability is particularly beneficial in medical imaging, where issues such as limited data availability and strict privacy regulations constrain the accessibility and scale of datasets. High-quality synthetic image generation can substantially improve data augmentation practices, enhance the robustness and generalizability of diagnostic algorithms, and support comprehensive simulation studies. Generative modeling techniques have progressively evolved, transitioning from classical latent variable approaches, exemplified by Variational Autoencoders (VAEs) introduced by Kingma and Welling [1], which encode data into continuous latent spaces for effective reconstruction and sampling. Following this, Generative Adversarial Networks (GANs), pioneered by Goodfellow et al. [2], have gained prominence through their ability to produce visually compelling images by training generators to deceive discriminative networks. More recently, Diffusion Models, as demonstrated by Ho et al. [3], have emerged as powerful alternatives capable of generating high-quality images through iterative denoising processes guided by probabilistic frameworks.

In this study, we evaluate these three distinct generative modeling paradigms—VAEs, GANs, and Diffusion Models—using the BloodMNIST dataset, comprising 28×28 RGB images of peripheral blood cells. This dataset poses specific challenges such as limited image resolution, nuanced inter-class distinctions, and complex cellular morphologies, rendering it an ideal benchmark for assessing the comparative strengths and limitations of generative models within biomedical contexts.

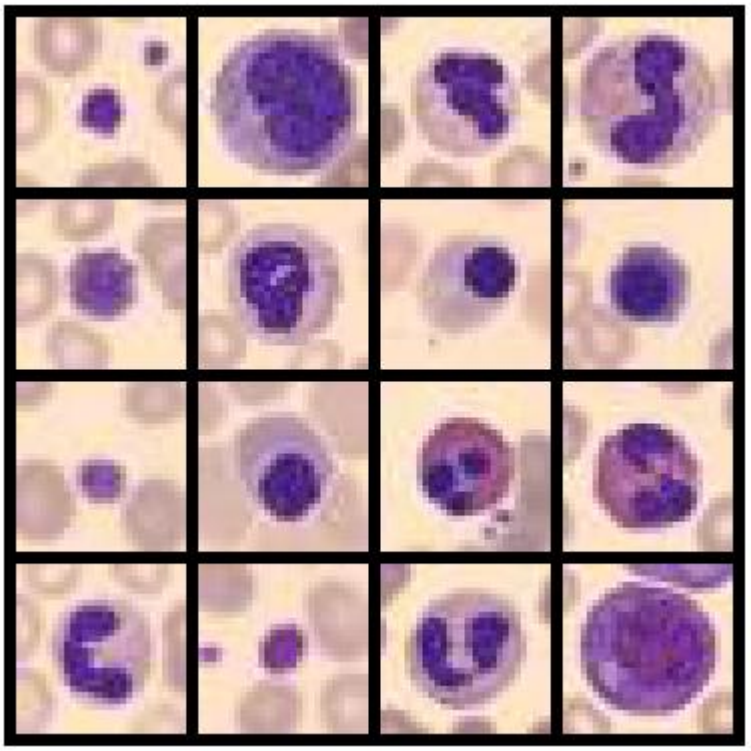


Fig. 1. Representative examples of peripheral blood cell images from the BloodMNIST training dataset illustrating the eight distinct cell types.

Description of the Problem

The BloodMNIST dataset [4] comprises microscopic images representing eight distinct classes of peripheral blood cells: neutrophils, eosinophils, basophils, lymphocytes, monocytes, immature granulocytes, erythroblasts, and platelets. Each cell class possesses unique morphological features and textures, critical for accurate clinical diagnosis and medical research. Effective recognition and synthesis of such cellular images have significant implications, exemplified by recent advancements where convolutional neural networks achieved human-level performance in diagnosing acute myeloid leukemia through microscopic cell images [5].

The primary objective of this research is to develop and evaluate generative models capable of accurately capturing the underlying distribution of blood cell images within the BloodMNIST dataset. The goal is to synthesize novel cell images that are statistically and visually indistinguishable from authentic samples, thus potentially facilitating data augmentation and robustness of diagnostic tools.

However, several inherent challenges complicate this task:

- **High intra-class variability:** Significant differences in cell morphology, staining patterns, and size occur within the same cell type, complicating accurate generative modeling.
- **Low inter-class contrast:** Certain blood cell classes display subtle morphological distinctions, creating difficulties in both recognition and generative tasks.
- **Limited image resolution:** The small image dimensions (28×28 pixels) restrict the information available per sample, consequently limiting the representational capacity of generative architectures.

Furthermore, to robustly quantify the quality and diversity of generated samples, a reliable evaluation metric is essential. The Fréchet Inception Distance (FID), introduced by Heusel et al. [6], is widely employed in generative modeling research. The FID metric assesses the similarity between real and generated image distributions using deep convolutional features extracted via a pretrained InceptionV3 network [7], effectively capturing both the fidelity and variability of synthetic images.

Approach

To tackle the problem of generating realistic blood cell images, we chose three well-established generative model types, each grounded in different principles of representation learning and synthesis.

First, the **Variational Autoencoder (VAE)** was implemented as a probabilistic encoder-decoder system. The encoder takes an input image and compresses it into a latent space characterized by a distribution (mean and variance), rather than a single point. This stochastic embedding encourages the model to learn a smooth, continuous latent space where sampling is possible. The decoder then reconstructs the image from this latent representation. Our architecture uses a series of convolutional layers to progressively reduce spatial dimensions and increase feature abstraction in the encoder, and mirrored transposed convolutions in the decoder to upsample back to image space. The training objective combines a reconstruction loss, which forces the output to resemble the input, and a Kullback-Leibler (KL) divergence term that regularizes the latent distribution towards a standard normal prior. This balance helps avoid overfitting and enables meaningful interpolation in latent space.

Next, the **Deep Convolutional GAN (DCGAN)** embraces an adversarial approach, where two networks—the generator and the discriminator—compete in a zero-sum game. The generator learns to produce synthetic images from random noise, while the discriminator attempts to distinguish these generated images from real ones. Through this interplay, the generator progressively improves its outputs to "fool" the discriminator. Our DCGAN generator starts with a low-dimensional noise vector, then uses a sequence of transpose convolutional layers with batch normalization and dropout for improved generalization and training stability. Dropout layers serve as a regularization mechanism, preventing the network from becoming overly confident and reducing overfitting. We carefully initialized the weights to ensure stable training. The discriminator, although not detailed here, is trained simultaneously to improve its real-vs-fake classification. This adversarial training paradigm often results in sharp, realistic images but can be challenging to stabilize.

Finally, the **Conditional Diffusion Model** adopts a fundamentally different strategy. It learns to generate images by gradually reversing a noising process—starting from pure noise and denoising step-by-step until a clean image emerges. This model uses a U-Net architecture composed of residual blocks with conditional normalization layers. These normalization layers adjust the feature maps based on a time embedding that encodes the current denoising step, allowing the network to adaptively learn how to clean images at various noise levels. The residual blocks and skip connections in the U-Net help preserve spatial information and enable deep feature extraction. By training on corrupted images paired with the noise added, the model learns to predict and remove noise iteratively. Although computationally heavier due to multiple steps of denoising during sampling, diffusion models have been shown to generate high-quality images with less mode collapse.

Together, these approaches allow us to compare different philosophies of generative modeling—latent space reconstruction, adversarial synthesis, and iterative denoising—on the same challenging dataset.

Experimental Setup

All models were trained on the full BloodMNIST dataset containing over 17,000 samples, resized to 28×28 pixels in RGB format. The latent dimension was fixed at 128 for both the VAE and DCGAN.

- **Training Parameters:**
  - VAE: Optimized using Adam with a learning rate of 1e-3, minimizing the combined reconstruction and KL loss.
  - DCGAN: Used Adam optimizer with a learning rate of 2e-4, incorporating spectral normalization and dropout to stabilize adversarial training.
  - Diffusion Model: Employed a U-Net backbone with sinusoidal time embeddings, trained with mean squared error loss on predicted noise.

Each model was trained for sufficient epochs until convergence, monitored by loss curves presented in Figure 2.

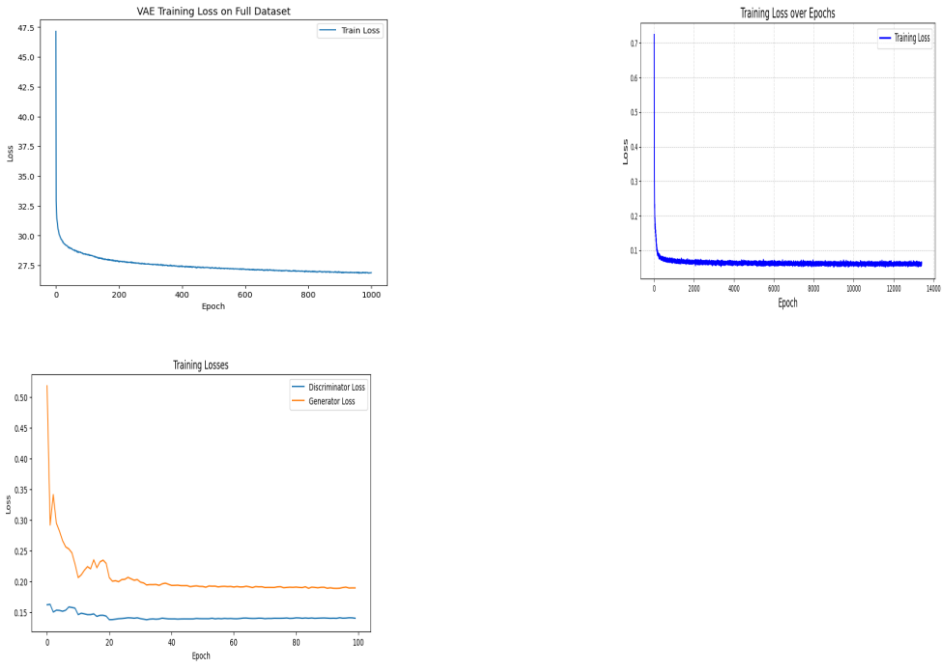


Fig. 2. Training loss curves for VAE, DCGAN, and Diffusion models. The VAE shows steady decrease in reconstruction and KL losses, DCGAN's adversarial loss stabilizes after initial fluctuations, and the diffusion model converges gradually over training steps.

For quantitative evaluation, we generated 10,000 synthetic images from each trained model, and computed the Fréchet Inception Distance (FID) against 10,000 randomly sampled real images. This process was repeated across multiple independent runs with varying random seeds to ensure robustness.

Analysis of the Results

After training all three models, we evaluated their performance primarily through the Fréchet Inception Distance (FID) metric, which quantifies how closely the distribution of generated images matches that of the real data in a high-level feature space. A lower FID indicates higher similarity and better generation quality.

The results clearly indicate that the DCGAN outperforms the other two models with an average FID of 96.88. This is not entirely surprising given that GANs, and DCGANs in particular, are well-known for their ability to generate visually sharp and diverse images. The adversarial setup forces the generator to capture intricate textures and fine details—such as the subtle color gradations and cellular structures in blood images—that other models may miss.

The diffusion model achieves the second-best score at 108.46. While slightly behind DCGAN, it still produces samples with convincing global structure and coherence. Diffusion models, due to their iterative refinement, tend to generate smoother images and maintain diversity without collapsing to a few modes. However, some of the fine textural details are less distinct compared to DCGAN outputs. This could be related to the limited number of denoising steps or model capacity constraints. Importantly, diffusion models offer a more stable training process and are less prone to the instability issues that often plague GAN training.

The VAE shows the highest FID score of 172.70, reflecting poorer visual quality in the generated images. This is expected because VAEs rely on pixel-wise reconstruction losses, which often produce blurrier results. The KL divergence regularization, while beneficial for latent space smoothness, imposes a constraint that can limit the model's ability to capture high-frequency details. Although the VAE excels at learning a meaningful latent representation and allows easy sampling, it sacrifices sharpness and fine textures, which are critical for medical image realism.

When we visually compared generated images with real samples, the DCGAN images stood out with crisp edges and realistic morphology closely mimicking true blood cells. The diffusion model samples were smooth, coherent, and consistent with general cell shapes, but sometimes lacked crisp boundaries. Meanwhile, VAE outputs appeared softer and somewhat fuzzy, failing to capture the subtle intricacies of cell structure.

Examining the training loss curves further complements these observations. The VAE loss steadily decreases, showing consistent improvement in reconstruction, but the output quality plateaus in terms of detail. The DCGAN loss oscillates due to the adversarial nature but stabilizes eventually, indicating a balance between generator and discriminator. The diffusion model loss slowly converges over many epochs, reflecting the gradual refinement process intrinsic to diffusion training.

Table 1: Average Fréchet Inception Distance (FID) scores evaluating generative model performance on BloodMNIST. Lower values indicate higher sample quality and diversity.

Model	**FID Score**
DCGAN	96.88
Diffusion	108.46
VAE	172.70

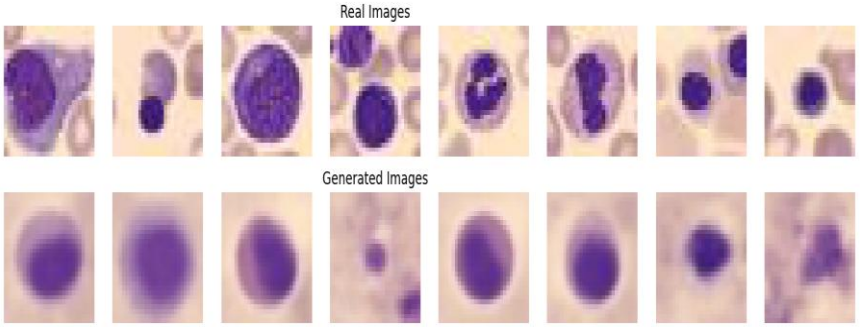


Fig. 3. Visual comparison of real blood cell images (top row) with generated samples (bottom row) of Variational Autoencoder.

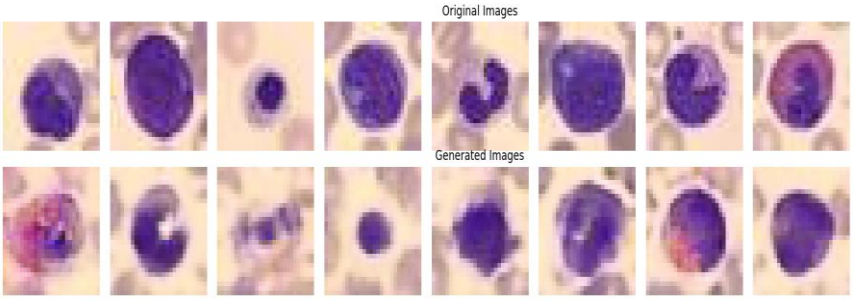


Fig. 4. Visual comparison of real blood cell images (top row) with generated samples (bottom row) of DC-GAN

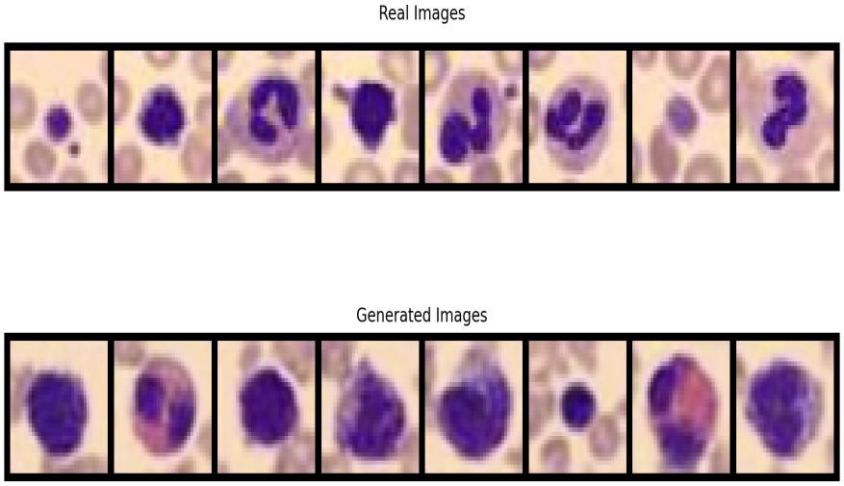


Fig. 5. Visual comparison of real blood cell images (top row) with generated samples (bottom row) of Diffusion Model.

In summary, while each model has strengths, the adversarial DCGAN emerges as the best at generating high-quality, diverse blood cell images on this dataset. The diffusion model is a promising alternative, balancing quality and stability, and the VAE provides a reliable, interpretable latent structure though at the expense of sample fidelity.

Conclusion

In summary, this project systematically evaluated three generative architectures for peripheral blood cell image synthesis using the BloodMNIST dataset. Our experiments reveal that DCGAN provides the best balance of image quality, diversity, and training efficiency, evidenced by its lowest FID score and visually compelling outputs. The diffusion model, while promising, demands higher computational resources and slightly underperforms DCGAN in this setting. VAEs offer a stable and interpretable latent representation but fall short in sample fidelity.

These insights inform the selection of generative methods for biomedical image augmentation and related applications. Future work could investigate hybrid models combining variational and adversarial elements or enhance diffusion sampling speed to achieve superior synthesis.

References:

[1] Kingma, D.P., Welling, M.: Auto-encoding variational Bayes. In: Proceedings of the 2nd International Conference on Learning Representations (ICLR), arXiv preprint arXiv:1312.6114 (2014).

[2] Goodfellow, I.J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative adversarial nets. In: Advances in Neural Information Processing Systems, vol. 27, pp. 2672–2680 (2014).

[3] Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. In: Advances in Neural Information Processing Systems, vol. 33, pp. 6840–6851 (2020).

[4] Yang, J., Shi, R., Ni, B.: MedMNIST classification decathlon: A lightweight AutoML benchmark for medical image analysis. In: Proceedings of IEEE International Symposium on Biomedical Imaging (ISBI), pp. 191–195 (2021).

[5] Matek, C., Schwarz, S., Spiekermann, K., Marr, C.: Human-level recognition of blast cells in acute myeloid leukaemia with convolutional neural networks. Nat. Mach. Intell. **1**, 538–544 (2019).

[6] Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In: Advances in Neural Information Processing Systems, vol. 30, pp. 6626–6637 (2017).

[7] Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z.: Rethinking the inception architecture for computer vision. In: Proceedings of IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2818–2826 (2016).